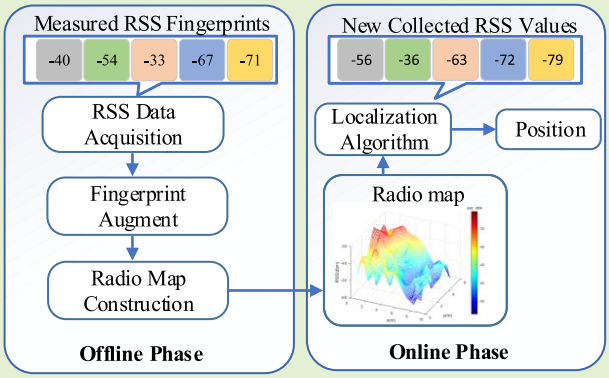
Fingerprint Augment Based on Super-Resolution for WiFi Fingerprint Based Indoor Localization

Tian Lan, Xianmin Wang, Zhikun Chen, Jinkang Zhu, *Life Member, IEEE*, and Sihai Zhang, *Senior Member, IEEE*

***Abstract*—WiFi fingerprint based indoor localization has become a key research direction in the field of indoor localization due to its high positioning accuracy and low equipment deployment cost. Increasing the number of reference point collected offline can improve the positioning accuracy, however it yields excessive cost of offline collection. Fingerprint augment is an effective solution to reduce the cost while ensuring the positioning accuracy. In this paper, we are pioneering to propose a fingerprint augment framework based on super-resolution (FASR), which achieves the fusion of fingerprint augment and super-resolution based on mutual conversion between fingerprint data and fingerprint image. The processing framework of FASR is formulated and the implementation of three modules in FASR are given, including Fingerprint-To-Image Conversion module, Super-Resolution module and Image-To-Fingerprint Conversion module. Simulated and real data experiments reveal the feasibility and effectiveness of the FASR. In addition, we explore the impact of two key engineering parameters on the performance of the FASR method. Our work demonstrates the new application of super-resolution in image processing field in wireless indoor localization topics.**



***Index Terms*—Indoor WiFi localization, fingerprint augment, super-resolution, WiFi fingerprint image.**

I. Introduction

WITH the development of mobile communication technology, the use of mobile device has become an important part of human's daily life. Various user-centric mobile applications emerge in endlessly many of which hope to provide smarter services based on users' real-time locations [1]. Global positioning system (GPS) is widely and reliably used in outdoor positioning and navigation services [2]. As for indoor positioning, however, due to the scattering, attenuation and multi-path propagation effects of wireless signals, positioning technologies such as GPS commonly used outdoors can not work well. Therefore, how to achieve low-cost and high-accuracy indoor localization has extremely high research value. Nowadays, indoor localization methods include WiFi [3], [4], Bluetooth [5], [6] and ZigBee [7], among which indoor WiFi localization has become research hot-spot due to its high positioning accuracy and low material cost [8].

Indoor WiFi localization can be divided into two categories comprising ranging-based WiFi positioning method [9] and fingerprint-based WiFi positioning method [10]. Ranging-based WiFi positioning uses information such as the angle of arrival (AOA) and time of arrival (TOA) to calculate the distance between the user equipment (UE) and access points (APs), then locates the user equipment based on the calculated distance. This method requires the location of APs to be prior known, and the positioning accuracy is affected by the ranging precision between UE and APs. Fingerprint-based positioning uses the received signal information of each AP collected by the UE at each reference point (RP) to construct fingerprint database, and then locates the user equipment by comparing the fingerprint data given by the UE at testing location with the fingerprint database.

In details, fingerprint-based WiFi positioning includes two phases, the offline phase and the online phase [10], [11]. The offline phase completes the perception of the positioning scene

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and the construction of the fingerprint database. Fingerprint data usually includes the coordinate of RP and received signal information of each AP collected at the RP. Since the received signal strength (RSS) of WiFi can be easily collected by WiFi module in mobile device, RSS is usually adopted as the received signal information. In the online phase, positioning is achieved by comparing the online fingerprint data (lacking the coordinate) with the fingerprint database collected in offline phase. Due to its high positioning accuracy, fingerprint-based WiFi positioning method has become a key research direction in indoor positioning scheme.

The accuracy of fingerprint-based WiFi positioning can be improved by increasing the number of RPs or the amount of fingerprint data collected at each RP in fingerprint database, which, however, yields excessive cost of offline acquisition [12]. Hence, how to reduce the cost of offline acquisition in such case is an important research problem. Existing solutions include RSS collection based on multi-user participation [13] and RSS estimation based on interpolation [14]. The former reduces the cost of single user by distributing the collection burden to multiple users, such as explicit crowdsourcing-based data collection [13], implicit data collection [15], and partially-labeled fingerprints [16]. The latter uses the fingerprint of each RP in fingerprint database to generate virtual fingerprint, which is called fingerprint augment (FA) [17]. FA methods can be divided into two categories: generating virtual fingerprints at each RP or generating virtual fingerprints at virtual RPs(VRPs). The former uses algorithms, such as GAN [18], to generate virtual fingerprints at each RP, thus reducing the acquisition cost at each RP, while the latter is more concerned with reducing the number of RPs needed in the whole localization scenario.The latter one has received more attention because the extra estimated VRPs help improve the positioning accuracy without excessive offline acquisition.

RSS estimation based on propagation model [19] or regression model [20] are commonly used fingerprint augment methods. The propagation models-based RSS estimation method estimates RSS value at VRPs to create virtual fingerprints through propagation models and sparse fingerprint database collected. However, it is difficult to find suitable propagation model in changeable environment and indoor layouts. The regression models-based RSS estimation uses the fingerprint data collected at RPs to calculate the regression parameters, and then estimates the RSS value at VRPs based on the regression model with known parameters. The limitation is that it is difficult to determine suitable regression model in complex indoor environment. Nowadays, how to find the optimal mapping function between the raw sparse fingerprint database and augmented dense fingerprint database poses a challenge to fingerprint augment.

In this paper, we devise a super-resolution based deep neural network (DNN) in image processing field to approximate the mapping function between sparse and dense fingerprint database. Single image super-resolution aims to reconstruct a high-resolution image from a single low-resolution image [21]–[23]. Recently, super-resolution based on DNN has achieved remarkable performance [24], due to its nonlinear mapping capability, which may well deal with the fingerprint augment process. In this paper, we propose a super-resolution based WiFi fingerprint augment framework to integrate the super-resolution with WiFi fingerprint augment. In this framework, three key components are designed, including WiFi fingerprint database to low-resolution images conversion, super-resolution for low-resolution images and high-resolution images to WiFi fingerprint database conversion.

The main contributions of this paper are as follows:

- We are pioneering to propose the Fingerprint Augment Based on Super-Resolution (FASR) framework for indoor WiFi fingerprint augment which consists of *Fingerprint-To-Image* for converting WiFi fingerprint database to images, super-resolution based on DNN for reconstructing the high-resolution fingerprint images from low-resolution fingerprint images and *Image-To-Fingerprint* for converting high-resolution images to fingerprint database.
- We clarify the effect of two engineering parameters, i.e., the geographic spacing of RPs and the upsampling factor of DNN, on the performance of the FASR through extensive experiments which provide practical guidance about how to choose these parameters in practice. We validate the performance improvement in fingerprint augment accuracy and positioning accuracy brought by the proposed FASR through the comparative experiments in both numeric simulation and test platform.

The remainder of this paper is organized as follows. Section II introduces some representative works on fingerprint augment and super-resolution. In Section III, we propose the framework of FASR to integrate the super-resolution into the field of WiFi fingerprint augment. Section IV introduces the implementations of three modules in FASR, including Fingerprint-To-Image conversion, Image-to-Fingerprint conversion and super-resolution for fingerprint image. Then, the fingerprint augment and localization performances of FASR are analyzed in Section V. Besides, the impact of two key engineering parameters is analyzed in Section VI. Finally, the conclusions of this paper are drawn in Section VII.

II. Related Works

In this section, we provide a brief introduction to fingerprint augment and image super-resolution.

A. Fingerprint Augment

Existing fingerprint augment methods include propagation models-based RSS estimation [19] and regression-based RSS estimation [20]. The propagation models-based RSS estimation method uses the indoor signal propagation models to estimate the RSS value in the line-of-sight (LOS) and non-line-of-sight (NLOS) areas. Commonly used propagation models include OneSlope model, Multiwall model, Motif model and MultiWall MultiFloor model [25], [26]. The propagation model's parameters of each AP are calculated through a small amount of RSS data collected, and then the propagation model with known parameters is used to estimate the RSS value at VRPs as fingerprint data.

The regression-based RSS estimation method uses regression model to estimate the RSS value at VRPs, where the

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parameters of the regression model could be calculated based on RSS data and position coordinates prior collected from a small number of RPs. Among all the regression-based methods, Gaussian process regression (GPR) [27] is one of the most commonly used regression model. The GPR based fingerprint augment method can estimate the Gaussian distribution obeyed by RSS value at VRP, and the choice of mean function and covariance function in GPR model is flexible. Researchers have been exploring the most suitable combination of mean function and covariance function for fingerprint augment [28], [29].

As one of the determinants of Gaussian process, the mean function plays an important role in GPR based fingerprint augment. The basic GPR (B-GPR) [30] method use zero mean to reduce the computational complexity. However, using zero mean for RSS estimation makes the estimated RSS on the VRP far away from the RP to tend to zero, which thereby reduces the accuracy of fingerprint augment. Therefore, it is necessary to find a non-zero mean function to improve the accuracy of fingerprint augment. Sun *et al.* [31] consider the real RSS propagation characteristics, and use the Log-Distance path loss model [32] as the mean function. However, in real environment, the attenuation of the actual signal may be heavily different from the results of the logarithmic distance path loss model, thus GPR based on this mean function can not obtain ideal performance. Zou *et al.* [20] propose the Winips scheme, which uses a complex two-dimensional polynomial model as the mean function. However, the large number of parameters of the mean function increased the computational complexity of GPR.

In addition, the choice of the covariance function is also important for the Gaussian process. Sun *et al.* [31] choose squared exponential kernel, Matern kernel [33], rational quadratic with isotropic distance measure kernel, noise kernel [34], constant kernel, linear kernel and compound kernels of these kernel functions as the covariance function respectively, and compare the performance through experiments. The experimental results show that Matern & Rational quadratic with isotropic distance measure compound kernel based GPR (MR-GPR) has good fingerprint augment performance.

B. Super-Resolution

Methods to solve the super-resolution problem are mainly divided into three categories: reconstruction-based super-resolution [35], interpolation-based super-resolution [36], and learning-based super-resolution [37]. Recently, with the progress of deep neural network research, learning-based super-resolution has been substantively developed.

Super-Resolution Convolutional Neural Network (SRCNN) is the first application of convolutional neural network in the field of super-resolution, which uses simple network structure to learn the mapping relationship between low-resolution and high-resolution images which exihits ideal super-resolution performance [38]. Fast Super-Resolution Convolutional Neural Networks (FSRCNN) increases the training speed of the network by 40 times and reduces the computational cost with better super-resolution performance by improving the network structure of SRCNN [39]. Efficient Sub-Pixel Convolutional Neural Network (ESPCN) directly performs convolution on low-resolution image to extract features and then use sub-pixel convolution layer to obtain high-resolution image, thus reduces the computational complexity of the network [21].

Convolutional Networks (VDSR) for super-resolution problem based on deep convolutional network improves the convergence speed during network training by introducing the residual network in Deep Residual Network (ResNet) [40], which solves the gradient explosion problem and obtain better restoration quality [37]. Deeply-Recursive Convolutional Network (DRCN) uses deep recursive convolutional network structure to improve the super-resolution performance, and then speeds up the convergence of the network by introducing the idea of residual learning [41]. Based on ResNet, VDSR and DRCN, Deep Recursive Residual Network (DRRN) with up to 52 convolutional layers uses residual learning based on global and local methods to reduce the difficulty of deep network training, and uses recursive learning to control the number of model parameters, which further improves the accuracy of super-resolution [42]. Super-Resolution Deep residual network (SRResNet) uses generative adversarial network and modifies the residual network in ResNet. The high-resolution images obtained by the method are relatively smooth and have ideal visual effects [43].

Enhanced Deep Super-Resolution Network (EDSR) further modifies the residual network used by SRResNet to extract more features in each layer, and obtain excellent performance in existing deep learning networks [22]. Residual Dense Network (RDN) proposes residual dense block (RDB) and global feature fusion (GFF) to extract and adaptively fuse features from all the layers in the LR space efficiently [23].

In this paper, we are pioneering to employ the deep neural networks, like SRCNN and EDSR, into fingerprint augment and devise a complete framework for fingerprint augment based on the intention of super-resolution. The advantage of the deep neural networks against the traditional propagation methods, i.e., propagation models-based and regression-based, is that deep neural networks have more powerful nonlinear mapping capability, which may bring better performance gain for wireless signal processing, if properly designed.

III. The FASR Framework

Fig. 1 shows the proposed fingerprint augment framework based on super-resolution (FASR), which consists of three modules. The *Fingerprint-To-Image Conversion* module converts the collected WiFi fingerprint database into low-resolution WiFi fingerprint images. The *Super-Resolution* module processes the low-resolution WiFi fingerprint image using deep neural network to obtain high-resolution image which realize the expansion of pixels. The *Image-To-Fingerprint Conversion* module transforms the high-resolution WiFi fingerprint images into the augmented WiFi fingerprint database. FASR receives the WiFi fingerprint data from the offline data collection phase, and the augmented fingerprint database will be directly used for localization.

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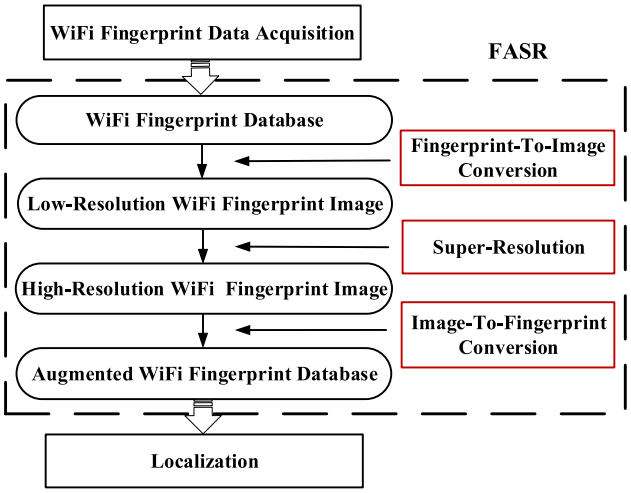

Fig. 1. The framework of FASR.

A. WiFi Fingerprint Database

Assume a typical indoor localization scenario with n APs denoted by $\{AP_1, AP_2, \ldots, AP_n\}$, m RPs divided by geographic distance λ is represented by $\{RP_1, RP_2, \ldots, RP_m\}$. $S_i = (x_i, y_i)$ is the two-dimension (2D) cartesian coordinate of the RP_i. Suppose the RSS value of AP_j measured at RP_i denoted by $\mathbf{R}_{i,j} = \{r^1_{i,j}, r^2_{i,j}, \ldots, r^T_{i,j}\}$, $r^t_{i,j}$ represents the RSS value of the collected t-th sample, $1 \leq t \leq T$. One WiFi fingerprint is formed at one RP, which is composed by the coordinate of the RP and RSS values of all APs collected at the RP, so the i-th WiFi fingerprint is represented by $\{S_i, \mathbf{R}_{i,1}, \mathbf{R}_{i,2}, \ldots, \mathbf{R}_{i,n}\}$. And the whole set of received WiFi fingerprint database in the offline phase can be described as:

$$\mathbf{R} = \begin{bmatrix} S_1 & \mathbf{R}_{1,1} & \mathbf{R}_{1,2} & \ldots & \mathbf{R}_{1,n} \\ S_2 & \mathbf{R}_{2,1} & \mathbf{R}_{2,2} & \ldots & \mathbf{R}_{2,n} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ S_m & \mathbf{R}_{m,1} & \mathbf{R}_{m,2} & \ldots & \mathbf{R}_{m,n} \end{bmatrix}. \tag{1}$$

Suppose there are m' VRPs denoted by $\{VRP_1, VRP_2, \ldots, VRP_{m'}\}$. Here we use the fingerprint database $\mathbf{R}$ with m sparse RPs for fingerprint augment to estimate the fingerprint data at m' VRPs. After performing fingerprint augment, the number of WiFi fingerprints is $M = m + m'$, and the WiFi fingerprint database is updated to:

$$\mathbf{R}^{\mathbf{FA}} = \begin{bmatrix} S_1 & \mathbf{R}_{1,1} & \mathbf{R}_{1,2} & \ldots & \mathbf{R}_{1,n} \\ S_2 & \mathbf{R}_{2,1} & \mathbf{R}_{2,2} & \ldots & \mathbf{R}_{2,n} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ S_m & \mathbf{R}_{m,1} & \mathbf{R}_{m,2} & \ldots & \mathbf{R}_{m,n} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ S_M & \mathbf{R}_{M,1} & \mathbf{R}_{M,2} & \ldots & \mathbf{R}_{M,n} \end{bmatrix}. \tag{2}$$

In FASR, $\mathbf{R}$ will be converted into low-resolution images for super-resolution, and the resulting high-resolution image will be reconstructed into $\mathbf{R}^{\mathbf{FA}}$.

B. Fingerprint-To-Image and Image-To-Fingerprint

In the FASR, we transform the fingerprint augment problem into a super-resolution problem of the fingerprint images based on the conversion of the WiFi fingerprint.

1) Fingerprint-To-Image Conversion: Suppose I_{LR} represents the low-resolution fingerprint images, then the conversion method f is defined as follows:

$$f : \mathbf{R} \rightarrow I_{LR}. \tag{3}$$

During the conversion of f, a RP in the fingerprint database corresponds to a pixel in the fingerprint image, and the RSS value of the RP corresponds to the pixel value of the corresponding pixel. The 2D coordinates of RPs can not be directly reflected in the fingerprint image, but the relative position of the RPs in indoor environment can be shown by the relative position of the pixels in image.

2) Image-To-Fingerprint Conversion: Suppose I_{HR} represents the high-resolution fingerprint images. The conversion method f' is equivalent to the inverse process of f, and is defined by:

$$f' : I_{HR} \rightarrow \mathbf{R}^{\mathbf{FA}}. \tag{4}$$

Same as f, one pixel corresponds to one RP, and the value of each pixel needs to be mapped back to RSS value. The coordinates of the newly generated RPs should be allocated according to the coordinates of original RPs. The obtained augmented fingerprint database achieves the density expansion of the original fingerprint database, and the geographic distance of RPs is reduced.

C. Super-Resolution

The super-resolution method f^{SR} is defined as follows:

$$f^{SR} : I_{LR} \rightarrow I_{HR}. \tag{5}$$

In the super-resolution process, the deep neural network first learns the mapping function between the low-resolution fingerprint image and the high-resolution fingerprint image based on the training data. After that, the given low-resolution fingerprint image I_{LR} is input into the well-trained network to obtain the required high-resolution fingerprint image I_{HR}.

IV. Implementation of FASR

In this section, the implementation of FASR is introduced.

A. Fingerprint-To-Image Conversion Method

Three processing are performed for this conversion.

First, we segment the collected fingerprint database in the time dimension and average the RSS values in $\mathbf{R}_{i,j}$ in each segment. $\overline{R^{t'}_{i,j}}$ is the average RSS value of the t'-th segment of the AP_j at the RP_i in fingerprint database:

$$\overline{R^{t'}_{i,j}} = \frac{1}{k} \sum_{t=(t'-1)k+1}^{t'k} r^t_{i,j}, t' = \{1, 2, \ldots, \frac{T}{k}\}. \tag{6}$$

where k denotes the segment length, t' is the number of segment and a total of $\frac{T}{k}$ segment are obtained.

Second, the average RSS values at each segment in the fingerprint database are mapped to the gray range of $[0 \sim 255]$

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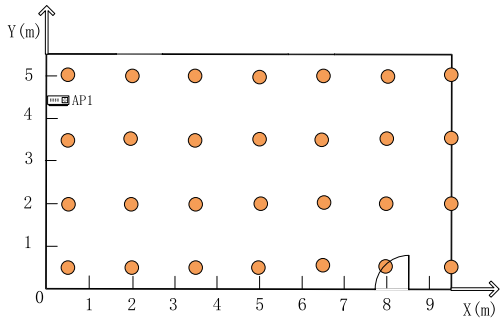
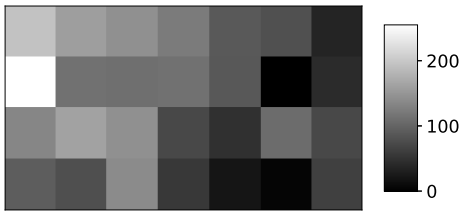


(a) RPs selected in a $9.5m \times 5.5m$ room.



(b) Fingerprint image with $\lambda_{LR} = 1.5m$.

Fig. 2. Fingerprint-To-Image conversion.

based on the following equation. $P_{i,j}^{t'}$ is the gray value of the pixel which is corresponding to the $\overline{R_{i,j}^{t'}}$ of RP_i:

$$P_{i,j}^{t'} = \frac{255(\overline{R_{i,j}^{t'}} - R_{min}^{j,t'})}{R_{max}^{j,t'} - R_{min}^{j,t'}}. \tag{7}$$

where $R_{max}^{j,t'}$ and $R_{min}^{j,t'}$ are the maximum and minimum average RSS values of AP_j at all RPs in the t'-th segment, respectively.

Finally, in a segment, for one AP, the gray values corresponding to RSS values are used to construct a gray image I_{LR} and the position of pixels in I_{LR} depend on the position of the corresponding RPs in the indoor environment. Therefore, for each time segment, we can obtain a total of n fingerprint images.

Note that the distribution of RPs must be grid shape and the geographic distance λ between any RPs must be recorded when generating a low-resolution fingerprint image. We define $\lambda_{LR} = \lambda$ in I_{LR} to record the λ in original fingerprint database and λ_{LR} will be used to reconstruct the fingerprint database from high-resolution fingerprint images.

Fig. 2 shows an example of the above method. We divide RPs (yellow dots) in a $9.5m \times 5.5m$ room with an AP and collect RSS to conduct fingerprint database in Fig. 2(a). We set $k = T$, and the converted I_{LR} with $\lambda_{LR} = 1.5m$ is drawn in Fig. 2(b).

B. Image-To-Fingerprint Conversion Method

This conversion method is also divided into three processing. First, the image gray values are mapped back to RSS values base on the following equation. $\overline{R_{i,j}^{t'}}'$ is the average RSS value of the t'-th segment mapped back:

$$\overline{R_{i,j}^{t'}}' = \frac{P_{i,j}^{t'}(R_{max}^{j,t'} - R_{min}^{j,t'})}{255} + R_{min}^{j,t'}. \tag{8}$$

where and the meanings of $P_{i,j}^{t'}$, $R_{max}^{j,t'}$ and $R_{min}^{j,t'}$ are the same as the Equation (7).

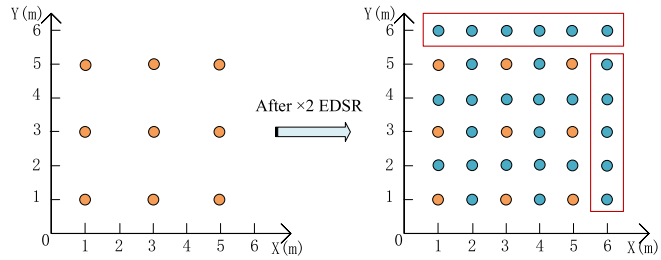


Fig. 3. Coordinates determination.

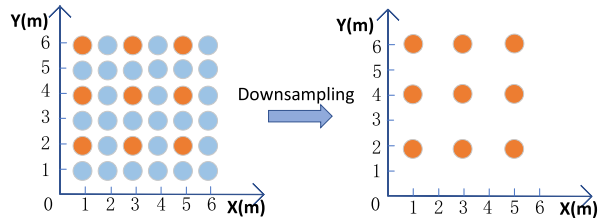


Fig. 4. Example of downsampling.

Second, we assign coordinates for newly generated RPs which correspond to the expanded pixels in I_{HR}, according to the coordinates of original RPs and λ_{HR}. The λ_{HR} means the geographic distance of RPs corresponding to the pixels in I_{HR} and is calculated by:

$$\lambda_{HR} = \frac{\lambda_{LR}}{\alpha}. \tag{9}$$

where λ_{LR} is the recorded geographic distance of RPs in I_{LR}, and α is the upsampling factor of the deep neural network. Furthermore, we crop the pixels at the edges of the fingerprint image according to the actual room size.

Finally, all fingerprint data are combined into a fingerprint database with the form of Equation (2).

We illustrate the determination of the coordinates of each pixel in I_{HR} and the cropping of fingerprint images in Fig. 3. There is an AP in a $5m \times 5\ m$ room, and the RSS values at 9 RPs are collected. A 3×3 low-resolution fingerprint image with $\lambda_{LR} = 2m$ is formed by the method in Section IV-A. Assume that we input this fingerprint image to a trained $\alpha = 2$ EDSR network to get a 6×6 fingerprint image with $\lambda_{HR} = 1m$. We first determine the coordinates of pixels (yellow dots) in the I_{HR} that originally existed in the I_{LR}, and then assign coordinate values to other pixels according to λ_{HR} of the I_{HR}. Due to the size of the room, the pixels at the top and right edges should be cropped.

C. Super-Resolution for Fingerprint Image

Existing deep neural network models for super-resolution [24] include SRCNN, FSRCNN, ESPCN, VDSR, DRRN, EDSR, RDN, which may all be adopted in the super-resolution task in FASR framework. In this paper, we choose the SRCNN and EDSR network to perform super-resolution for fingerprint images.

1) The Architecture of SRCNN Network: The network architecture of SRCNN includes 3 convolution layers. It firstly upscales the low-resolution image to the desired size using bicubic interpolation, then inputs it into SRCNN, and finally gets the high-resolution image from the output of the network.

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2) The Architecture of EDSR Neural Network: The network architecture of EDSR includes 256 feature channels for convolution layer and 32 residual blocks. EDSR performs feature extraction on the input low-resolution image, and finally outputs the high-resolution image through the upsampling layer. The upsampling layers for upsampling factor are $\times 2$, $\times 3$, and $\times 4$.

3) Downsampling Rule: According to the rule for generating VRPs in the fingerprint augment process, low-resolution fingerprint images need to be downsampled from high-resolution fingerprint images according to the following rule. For the super-resolution network with upsampling factor of α, one pixel is extracted every $\alpha - 1$ pixel from the edge of the image until all the optional points are selected.

Fig. 4 shows an example of downsampling when $\alpha = 2$.

V. Performance Evaluation

In this section, the proposed FASR is verified in simulated and real positioning scenarios and performances of fingerprint augment and localization are presented and discussed. The EDSR network with upsampling factor $\alpha = 3$ is adopted in experiment for super-resolution in both simulation and real scenarios. For comparison, B-GPR [30] and MR-GPR [31] are used for fingerprint augment in simulation scenario and real scenario, respectively. The weighted k-nearest neighbor (WKNN) localization algorithm [29] is adopted to carry out the localization in this work. Note that after the super-resolution operations, the augmented images are converted into RSS fingerprints, so the WKNN algorithm can be directly utilized for localization without any changes.

A. Experimental Data

1) Simulation Scenario: The room size of simulated scenario is $8.0m \times 8.0m \times 3m$, and 8 APs are installed around the positioning scenario. A total of 225 RPs are divided with $\lambda = 0.5m$.

The time invariant multipath channel is adopted to obtain the experimental data in the simulation positioning scenario [44]. The RSS value of any AP is generated by ray tracing technology [45]. For each AP, the RSS value at each location is obtained by superposing the signal from a direct path and six signals reflected from the walls. The Log-Distance path loss model in free space is used to calculate the RSS of each path.

Massive images are necessary for training the SR network, so we generated fingerprint data with $T = 3{,}000$ through simulation. Because eight APs are placed in positioning scenario, we can acquire eight fingerprint images based on Fingerprint-To-Image conversion method from each segment and a total of $(3000 \div k \times 8)$ fingerprint images with $\lambda_{HR} = 0.5m$ can be obtained as training dataset.

We select the newly generated fingerprint data of $\lambda = 0.5\ m$ with $T = 100$ to form the test dataset. For each segment, the RPs are extracted with $\lambda = 1.5m$ to construct the sparse WiFi fingerprint database which is converted into seven low-resolution fingerprint images with $\lambda_{LR} = 1.5\ m$, for super-resolution as I_{LR}. The original fingerprint data with $\lambda = 0.5\ m$ is used to evaluate the performance of fingerprint augment.

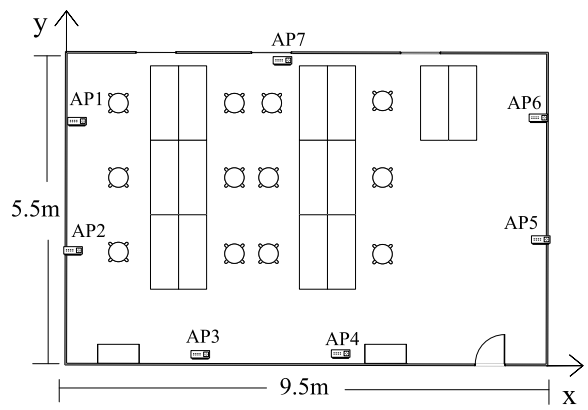

Fig. 5. The layout of the simulated and real positioning scenarios.

(a) AP. (b) Mobile phone holder.

Fig. 6. Data collection tools in real scenario.

In addition, 1,000 positioning test points are randomly selected to evaluate the localization performance of WiFi fingerprint database.

2) Real Scenario: The room size of real scenario is $5.5m \times 9.5m \times 3m$, and 7 APs with fixed locations are installed in each positioning scenario, as shown in Fig. 5. A total of 209 RPs are divided with $\lambda = 0.5m$. Fig. 6 shows the AP and the mobile phone holder used for RSS collection in the real scenario. Seven TL-WDR5620 APs are deployed around the room with the height of $1m$. By Android based RSS collection application we developed, RSS can be sampled every second. However, although the room is not very large which enables the RSS collection easier, the RSS value of an RP may be unable to be collected at some time due to the fluctuation of WiFi. So we use the RSS value of the previous period to replace the missing value for pre-filtering.

Totally 3,100 samples at each RP are collected. We select the fingerprint data of the first 3,000 samples to construct the training dataset. Same as the simulated scenario, a total of $(3000 \div k \times 7)$ fingerprint images with $\lambda_{HR} = 0.5m$ can be obtained.

The remaining fingerprint data with 100 samples are used to construct the test dataset. For each segment, we extract RPs with $\lambda = 1.5\ m$ from the collected fingerprint data to construct seven low-resolution fingerprint images with $\lambda_{LR} = 1.5\ m$ for super-resolution as I_{LR}. Before using the WKNN localization algorithm, we do mean filtering on the RSS of each AP at each RP, i.e., the final fingerprint database is the mean of the test dataset. The original fingerprint database containing 209 RPs are used to analyze the performance of fingerprint augment.

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TABLE I
LEARNING MODULE PARAMETERS OF SRCNN

Number of convolutional layers	3
Optimizer	Adam
Loss Function	MAE
Learning Rate	0.001
Batch Size	100
Kernel Size	3×3
Epoch	20

TABLE II
LEARNING MODULE PARAMETERS OF EDSR

Number of Residual Blocks	32
Number of Feature Maps	256
Optimizer	Adam
Loss Function	MAE
Learning Rate	0.001
Batch Size	100
Kernel Size	3×3
Epoch	20

Meanwhile, 38 positioning test points other than these 209 RPs are randomly selected to gather RSS used in the positioning phase, and 500 samples are collected at each positioning test location.

B. Neural Network Training

In our experiments, the major network parameter settings in SRCNN and EDSR are listed in Table I and Table II. Moreover, the upsampling factor of the FASR depends on the needs of the actual application and is determined before training the SR network.

C. Performance of Fingerprint Augment

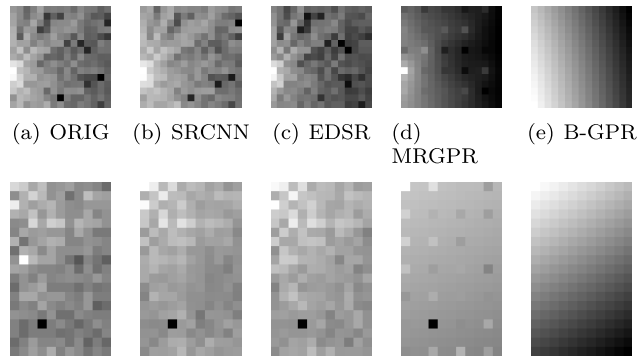
This part uses the experimental data in simulated and real scenario to explore the fingerprint augment performance of the FASR method and the two comparison methods. The test set contains fingerprint data of 100 segments, so we separately calculate the average error of fingerprint augment at each segment. The fingerprint augment error $e_{t'}^{FA}$ of t'-th segment is defined as follows:

$$e_{t'}^{FA} = \frac{1}{n \times m'} \sum_{j=1}^{n} \sum_{i=1}^{m'} |\overline{R_{i,j}^{t'}} - \overline{R_{i,j}^{t'}}'| \tag{10}$$

where m' is the numbers of RP after fingerprint augment.

For the FASR framework, the value of k affects the number of segments, which in turn affects the number of fingerprint images in training dataset. We first explore the influence of different k values on the fingerprint augment performance of FASR in the real scenario. The number of fingerprint data collected at each RP in the training dataset is 3,000 and the number of fingerprint data collected in the test set is 100. $k = 1, 2, 3, 4, 5, 6$ are selected for fingerprint image generation operation. Corresponding to the value of k, the number of fingerprint images (N) in the training dataset are 21,000, 10,500, 7,000, 5,250, 4,200, and 3,500, respectively.

Fig. 7 visually shows the effect of fingerprint images using FASR, B-GPR and MR-GPR in simulated and real scenario.



(a) ORIG (b) SRCNN (c) EDSR (d) MRGPR (e) B-GPR

(f) ORIG (g) SRCNN (h) EDSR (i) MRGPR (j) B-GPR

Fig. 7. Image comparison of fingerprint augment. (a)-(e) is in simulation scenario, and (f)-(j) is in real scenario. 'ORIG' means the high-resolution fingerprint image generated from the original data. The others means the high-resolution images generated of the augmented fingerprint database using FASR, B-GPR and MR-GPR.

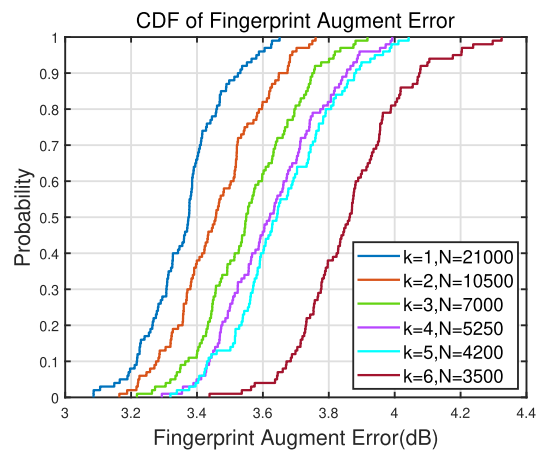


Fig. 8. Fingerprint augment error comparison versus k.

TABLE III
COMPARISON OF FINGERPRINT AUGMENT ERRORS(dB)

k	1	2	3	4	5	6
Error	3.36	3.46	3.56	3.63	3.66	3.87

Compared with MR-GPR and B-GPR, the fingerprint image obtained by FASR method is closer to the non-smooth feature of original fingerprint image.

Then we use the fingerprint images obtained under different k values to train the network, and perform fingerprint augment on the test dataset. The cumulative distributive function (CDF) of the fingerprint augment error at different values of k in real scenario is shown in Fig. 8. As we can see, the increase of k value leads to the increase of fingerprint augment error. And the average fingerprint augment errors of test dataset with different k values are calculated and shown in Tabel III. It can be clearly seen that the fingerprint augment performance obtained when $k = 1$ is the best. Therefore, in order to obtain good fingerprint augment performance, $k = 1$ is selected by default.

Fig. 9 shows the error of fingerprint augment using FASR ($k = 1$), B-GPR, MR-GPR in simulated and real scenario. Compared with MR-GPR and B-GPR, FASR has smallest errors of fingerprint augment. Meanwhile, the experimental results show that B-GPR is not robust, while the performances of FASR and MR-GPR are more stable.

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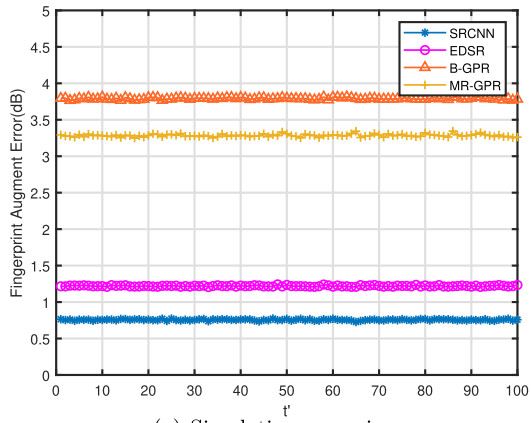
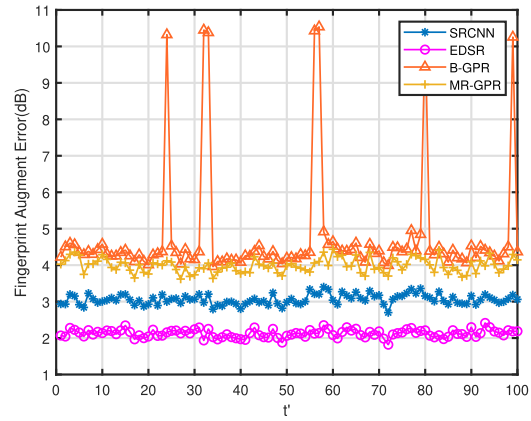


(a) Simulation scenario.



(b) Real scenario.

Fig. 9. Performance comparison of fingerprint augment.

The average error of fingerprint augment at 100 segments are calculated in Table IV. In simulation scenario, the proposed FASR outperforms MR-GPR and B-GPR. The fingerprint augment error of SRCNN is 79.9%, 76.8% lower than B-GPR and MR-GPR, while the fingerprint augment error of EDSR is 67.8%, 62.8% lower than B-GPR and MR-GPR, respectively. As for the real scenario, fingerprint augment error of FASR is also lower comparing to B-GPR and MR-GPR. Specifically, the fingerprint augment error of SRCNN is 37.6%, 23.3% lower than B-GPR and MR-GPR, while the fingerprint augment error of EDSR is 55.2%, 46.6% lower than B-GPR and MR-GPR. The experimental results validate the feasibility of FASR by its stable and outstanding fingerprint augment performance.

Furthermore, we observe that in the simulation scenario, SRCNN has a lower fingerprint augment error than EDSR, but the opposite trend in real scenario. This might be caused by the difference of the ability to extract features between two networks. The EDSR network uses the residual network to extract more features in each layer, so it can get better augment performance in complex real scenario. On the contrary, in the simulation scenario, the difference between data is relatively small, so the EDSR network is prone to overfittting, which makes it underperforms SRCNN.

D. Performance of Localization

This part explores the localization performance of fingerprint database obtained by the FASR ($k = 1$) method and the two comparison methods. The localization error e_i^L of i-th positioning test points is defined as follows:

$$e_i^L = \|S_i - S_i'\|_2 \tag{11}$$

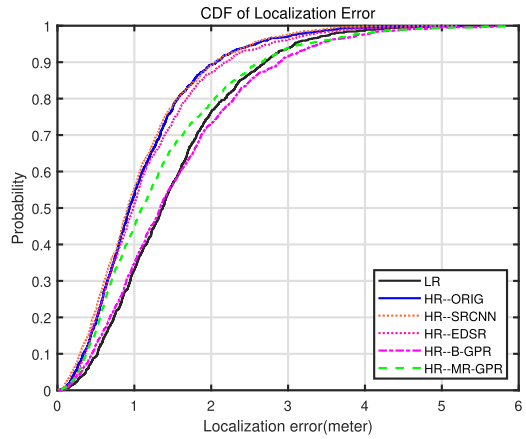
where S_i is the real physical position, S_i' is the estimation location.

The CDF of the localization error is drawn in Fig. 10. We found that the localization performance of FASR is closest to 'HR–ORIG' in different scenarios, which is better than B-GPR and MR-GPR. Moreover, the localization performance after fingerprint augment is better than 'LR' in the two scenarios, indicating the effectiveness of fingerprint augment.

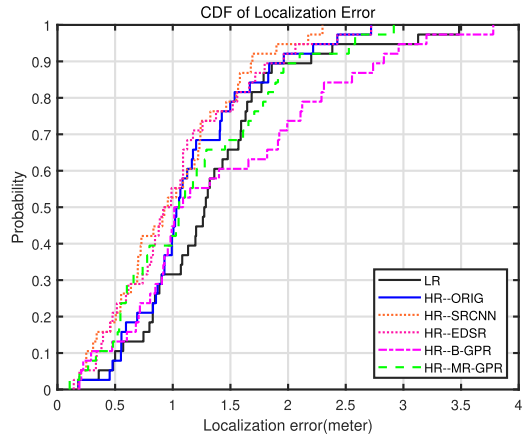
The average localization error is calculated in Tabel V. In simulation scenario, FASR performs better than MR-GPR and B-GPR. The average localization error of SRCNN is 26.7%, 20.1% lower than B-GPR and MR-GPR, while the

TABLE IV
COMPARISON OF FINGERPRINT AUGMENT ERRORS(dB)

Scenarios	SRCNN	EDSR	MR-GPR	B-GPR
Simulated	0.76	1.22	3.28	3.79
Real	3.06	2.13	3.99	4.75



(a) Simulation scenario.



(b) Real scenario.

Fig. 10. Performance comparison of localization. (1) 'LR' means the positioning result without fingerprint augment. 'HR–ORIG' means the original dense fingerprint database obtained by simulation or actual collection. The others represent the localization results of the augmented fingerprint database which is constructed by FASR-EDSR, B-GPR and MR-GPR, respectively. (2) For the two scenarios, the fingerprint database of 'LR' contains 28 fingerprints, the rest contain 209 fingerprints.

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TABLE V
COMPARISON OF LOCALIZATION ERRORS(m)

Scenarios	LR	ORIG	SRCNN	EDSR	MR-GPR	B-GPR
Simulated	1.49	1.11	1.07	1.16	1.34	1.46
Real	1.33	1.16	1.00	1.05	1.18	1.44

TABLE VI
THE SIMULATION ROOM SIZES OF DIFFERENT s_{pre}

$s_{pre}(m)$	0.6	0.9	1.5	3	6	9	12	15
Room length(m)	3.2	4.8	8	16	34	47	64	80
Room width(m)	3.2	4.8	8	16	34	47	64	80
Room height(m)	3	3	3	3	3	3	3	3

average localization error of EDSR is 20.5%, 12.3% lower than B-GPR and MR-GPR, respectively. As for the real scenario, the average localization error of FASR is also lower comparing to B-GPR and MR-GPR. Furthermore, the average localization error of SRCNN is 30.6%, 15.3% lower than B-GPR and MR-GPR, while the average localization error of EDSR is 27.1%, 5.0% lower than B-GPR and MR-GPR.

Moreover, we find that in the real scenario, SRCNN outperforms EDSR, and both of them outperform the original fingerprint database in average. This might be caused by large fluctuations in the fingerprint database in real scenario. Because it takes a long time to collect 209 RPs' fingerprint data in real scenario, the fingerprint database might have changed when collecting 38 positioning test points, which results in ORIG's performance in localization is not so good. We guess that the fingerprint database processed by SRCNN and EDSR could be closer to the real-time database, so the FASR could outperform ORIG. This also answers why EDSR is stronger than SRCNN in terms of augment performance but weaker than SRCNN in terms of localization performance in real scenario.

VI. IMPACT OF KEY PARAMETERS

This section explores the impact of two key engineering parameters, the geographic spacing of RPs (s_{pre}) and upsampling factor of DNN (α), in the offline phase of the WiFi fingerprint based indoor localization on the performance of the FASR method. The geographic spacing of RPs represents the spacing between RPs in the original collected sparse fingerprint database and the spacing between RPs after fingerprint augment is s_{pre}/α. We choose s_{pre} and α as independent variables and conduct experiments through the control variate method. Because the training time of SRCNN is much less than EDSR, we choose SRCNN as Super-Resolution Network in FASR model for the following experiments.

A. Experimental Data

In this part, the values of α may take 2, 3, 4, 5, 6 and s_{pre} may take 0.6m, 0.9m, 1.5m, 3m, 6m, 9m, 12m, 15m. So there are totally $5 \times 8 = 40$ experiment results including the fingerprint augment and localization.

Similar to the data acquisition method of the simulation scenario in Section V, 8 AP are installed around the positioning scenario. For different s_{pre}, we set different simulation room sizes, as shown in Table VI, which aims to investigate the performance of FASR method in rooms of different sizes. We generated fingerprint data with $T = 3,000$ through ray tracing technology and set $k = 1$. Based on Fingerprint-To-Image conversion method, total of 24,000 fingerprint images with $\lambda_{HR} = s_{pre}/\alpha$ can be obtained and used as training dataset.

In addition, we generated fingerprint data of $\lambda = s_{pre}/\alpha$ with $T = 100$ and set $k = 1$ to construct 700 low-resolution images I_{LR} with $\lambda_{LR} = s_{pre}$ as the test dataset.

Besides, 1,000 positioning test points are randomly selected other than RPs to perform the localization experiment.

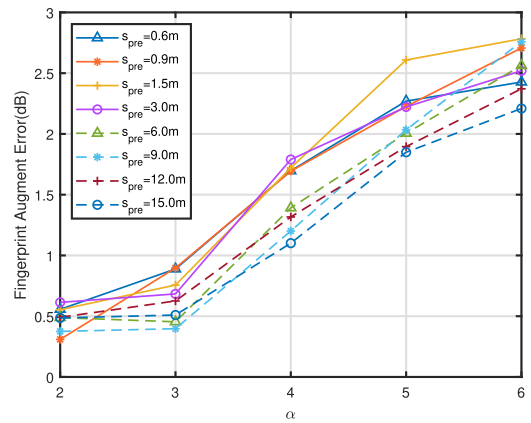

Fig. 11. Fingerprint augment performance comparison versus α and s_{pre} using SRCNN.

B. Performance Analysis Versus α and s_{pre}

Fig. 11 shows the fingerprint augment error using FASR with variable α and s_{pre}. With the increase of α, the fingerprint augment error shows an upward trend. In addition, the fingerprint augment performance when $s_{pre} > 3$ is more stable and better than the performance when $s_{pre} \leq 3$. In our experiments, $s_{pre} = 15$ has the best fingerprint augment performance.

Fig. 12 shows the localization performance of the fingerprint database after augmented by the FASR with the case of $s_{pre} = 0.6m, 0.9m, 1.5m, 3m$ in Fig.12(a) and the case of $s_{pre} = 6m, 9m, 12m, 15m$ in Fig.12(b). With the increase of α, the localization error gradually decrease. And when $s_{pre} = 3m, 6m, 9m, 12m$, the localization performance of the fingerprint database after fingerprint augment is more stable, approaching the performance of the original fingerprint database. In addition, for the original fingerprint database with any s_{pre}, the more the number of fingerprints, the better the localization performance.

According to the above observations, the impact of α and s_{pre} are summarized as follows:

For α, the fingerprint augment error gradually increases with the increase of α, but the localization error decreases with the increase of α. The main reason is that although increasing the intensity of fingerprint augment will increase the error of fingerprint augment, it also increases the number of fingerprints, which further improves the localization performance. Therefore, when the FASR method is used for fingerprint augment, the value of α can be increased according to the

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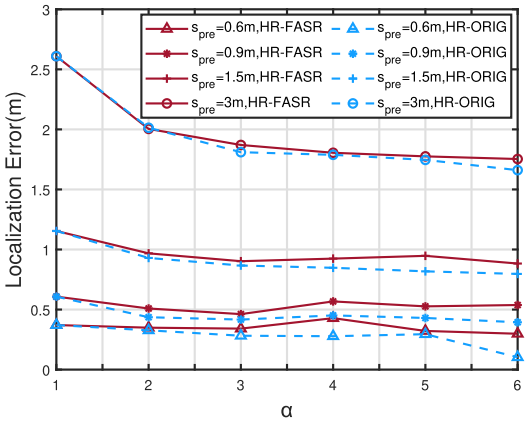
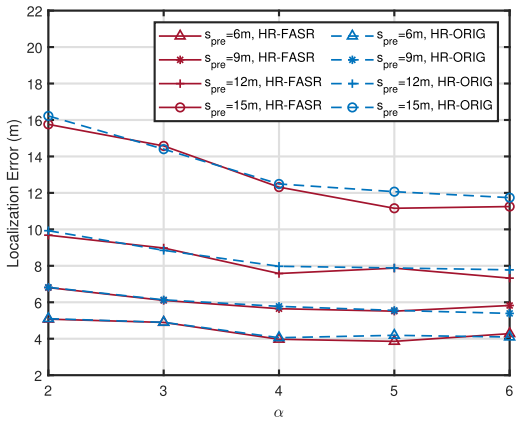


(a) $s_{pre} = 0.6m, 0.9m, 1.5m, 3m.$



(b) $s_{pre} = 6m, 9m, 12m, 15m.$

Fig. 12. Localization performance comparison versus α and s_{pre} using SRCNN.

actual positioning accuracy requirements. But the upper limit of α remains to be discussed.

For s_{pre}, the fingerprint augment error keeps a slight fluctuation with the increase of s_{pre}. In addition, when $s_{pre} = 0.6m, 0.9m, 1.5m$, the localization performance after fingerprint augment have large fluctuations than $s_{pre} = 3m, 6m, 9m, 12m, 15m$. Therefore, in order to maximize the benefit of fingerprint augment, the value of s_{pre} should be greater than $3m$. Meanwhile, a bigger s_{pre} corresponds to a larger room size. So it can be seen that the fingerprint augment performance of FASR is similar in different room size.

VII. Conclusion

In order to improve the accuracy of WiFi fingerprint-based indoor localization with limited fingerprints while reducing the site survey effort, this paper presents a FASR framework, which convert the fingerprint augment problem into super-resolution of fingerprint image. Furthermore, Fingerprint-To-Image conversion method and Image-To-Fingerprint conversion method are designed to implement the FASR. In order to evaluate the performance of FASR, we conducted experiments in two positioning scenarios, and compared the accuracy of RSS estimation and positioning with B-GPR and MR-GPR. Experiment results confirm potential of the FASR on RSS estimation as well as positioning accuracy. Moreover, we empirically demonstrate the impact about two engineering parameters, the geographic spacing of RPs and upsampling factor of DNN, on fingerprint augment and localization performances of the FASR method.

However, the performance of FASR has a close relationship with a lot of parameters, like different room sizes, different distributions of RPs, or different channel characteristics. In addition, our proposed FASR is validated in grid shape RPs only, not in non-grid shape RPs, which is also common in practical applications. In the future work, we plan to refine the FASR method to address these problems.

References

[1] Y. Gu, A. Lo, and I. Niemegeers, "A survey of indoor positioning systems for wireless personal networks," *IEEE Commun. Surveys Tuts.*, vol. 11, no. 1, pp. 13–32, Mar. 2009.

[2] M. Glocker, H. Landau, R. Leandro, and M. Nitschke, "Global precise multi-GNSS positioning with trimble centerpoint RTX," in *Proc. 6th ESA Workshop Satell. Navigat. Technol. Eur. Workshop GNSS Signals Signal Process.*, Dec. 2012, pp. 1–8.

[3] X. Ge and Z. Qu, "Optimization WiFi indoor positioning KNN algorithm location-based fingerprint," in *Proc. 7th IEEE Int. Conf. Softw. Eng. Sci. (ICSESS)*, Aug. 2016, pp. 135–137.

[4] Y. Wang and L. Shao, "Understanding occupancy pattern and improving building energy efficiency through Wi-Fi based indoor positioning," *Building Environ.*, vol. 114, pp. 106–117, Mar. 2017.

[5] M. Castillo-Cara, J. Lovón-Melgarejo, G. Bravo-Rocca, L. Orozco-Barbosa, and I. García-Varea, "An empirical study of the transmission power setting for Bluetooth-based indoor localization mechanisms," *Sensors*, vol. 17, no. 6, p. 1318, 2017.

[6] S. Liu, Y. Jiang, and A. Striegel, "Face-to-face proximity estimation using Bluetooth on smartphones," *IEEE Trans. Mobile Comput.*, vol. 13, no. 4, pp. 811–823, Apr. 2014.

[7] X. Liu, H. Guo, M. Yu, and Y. Li, "Improved ZigBee fingerprint method for indoor positioning," in *Proc. Pacific PNT Meeting*, 2013, pp. 616–623.

[8] K. Lin, M. Chen, J. Deng, M. M. Hassan, and G. Fortino, "Enhanced fingerprinting and trajectory prediction for IoT localization in smart buildings," *IEEE Trans. Autom. Sci. Eng.*, vol. 13, no. 3, pp. 1294–1307, Jul. 2016.

[9] J. Gjengset, J. Xiong, G. McPhillips, and K. Jamieson, "Phaser: Enabling phased array signal processing on commodity WiFi access points," in *Proc. 20th Annu. Int. Conf. Mobile Comput. Netw.*, Sep. 2014, pp. 153–164.

[10] P. Bahl and V. N. Padmanabhan, "RADAR: An in-building RF-based user location and tracking system," in *Proc. Conf. Comput. Commun., 19th Annu. Conf. IEEE Comput. Commun. Societies*, vol. 2, Mar. 2000, pp. 775–784.

[11] M. A. Youssef, A. Agrawala, and A. U. Shankar, "WLAN location determination via clustering and probability distributions," in *Proc. 1st IEEE Int. Conf. Pervasive Comput. Commun.*, Mar. 20003, pp. 143–150.

[12] J. Xiao, Z. Zhou, Y. Yi, and L. M. Ni, "A survey on wireless indoor localization from the device perspective," *ACM Comput. Surv.*, vol. 49, no. 2, p. 25, Oct. 2016.

[13] J. Ledlie *et al.*, "Molé: A scalable, user-generated WiFi positioning engine," *J. Location Based Services*, vol. 6, no. 2, pp. 55–80, 2012.

[14] G. Caso, L. De Nardis, and M. Di Benedetto, "Low-complexity offline and online strategies for Wi-Fi fingerprinting indoor positioning systems," in *Geographical and Fingerprinting Data to Create Systems for Indoor Positioning and Indoor/Outdoor Navigation*. New York, NY, USA: Academic, 2019, pp. 129–145.

[15] C. Luo, H. Hong, and M. C. Chan, "PiLoc: A self-calibrating participatory indoor localization system," in *Proc. 13th Int. Symp. Inf. Process. Sensor Netw. (IPSN)*, Apr. 2014, pp. 143–153.

[16] S. Sorour, Y. Lostanlen, S. Valaee, and K. Majeed, "Joint indoor localization and radio map construction with limited deployment load," *IEEE Trans. Mobile Comput.*, vol. 14, no. 5, pp. 1031–1043, May 2014.

[17] N. Hernández, M. Ocaña, J. M. Alonso, and E. Kim, "Continuous space estimation: Increasing WiFi-based indoor localization resolution without increasing the site-survey effort," *Sensors*, vol. 17, no. 1, p. 147, Jan. 2017.

[18] Q. Li *et al.*, "AF-DCGAN: Amplitude feature deep convolutional GAN for fingerprint construction in indoor localization systems," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 5, no. 3, pp. 468–480, Jun. 2021.

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[19] G. Caso and L. De Nardis, "On the applicability of multi-wall multi-floor propagation models to WiFi fingerprinting indoor positioning," in *Future Access Enablers of Ubiquitous and Intelligent Infrastructures*. Cham, Switzerland: Springer, 2015, pp. 166–172.

[20] H. Zou, M. Jin, H. Jiang, L. Xie, and C. J. Spanos, "WinIPS: WiFi-based non-intrusive indoor positioning system with online radio map construction and adaptation," *IEEE Trans. Wireless Commun.*, vol. 16, no. 12, pp. 8118–8130, Dec. 2017.

[21] W. Shi *et al.*, "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1874–1883.

[22] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 136–144.

[23] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[24] Z. Wang, J. Chen, and S. Hoi, "Deep learning for image super-resolution: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3365–3387, Oct. 2020.

[25] Widyawan, M. Klepal, and D. Pesch, "Influence of predicted and measured fingerprint on the accuracy of RSSI-based indoor location systems," in *Proc. 4th Workshop Positioning, Navigat. Commun.*, Mar. 2007, pp. 145–151.

[26] A. Borrelli, C. Monti, M. Vari, and F. Mazzenga, "Channel models for IEEE 802.11b indoor system design," in *Proc. IEEE Int. Conf. Commun.*, Jun. 2004, pp. 3701–3705.

[27] C. Rasmussen, "Gaussian processes in machine learning," in *Summer School on Machine Learning*. Berlin, Germany: Springer, 2003, pp. 63–71.

[28] J. Zhao, X. Gao, X. Wang, C. Li, M. Song, and Q. Sun, "An efficient radio map updating algorithm based on K-means and Gaussian process regression," *J. Navigat.*, vol. 71, no. 5, pp. 1055–1068, Sep. 2018.

[29] X. Wang, T. Lan, S. Zhang, and J. Zhu, "Signal distribution oriented mean functions in GPR based fingerprint augment," in *Proc. 11th Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Oct. 2019, pp. 1–6.

[30] S. Yiu, M. Dashti, H. Claussen, and F. Perez-Cruz, "Wireless RSSI fingerprinting localization," *Signal Process.*, vol. 131, pp. 235–244, Feb. 2017.

[31] W. Sun, M. Xue, H. Yu, H. Tang, and A. Lin, "Augmentation of fingerprints for indoor WiFi localization based on Gaussian process regression," *IEEE Trans. Veh. Technol.*, vol. 67, no. 11, pp. 10896–10905, Sep. 2018.

[32] G. Maral, M. Bousquet, and Z. Sun, *Satellite Communications Systems: Systems, Techniques and Technology*. Hoboken, NJ, USA: Wiley, 2020.

[33] T. Hofmann, B. Schölkopf, and A. J. Smola, "Kernel methods in machine learning," *Ann. Statist.*, vol. 36, no. 3, pp. 1171–1220, 2008.

[34] M. G. Genton, "Classes of kernels for machine learning: A statistics perspective," *J. Mach. Learn. Res.*, vol. 2, pp. 299–312, Dec. 2001.

[35] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[36] J. Yang, Z. Lin, and S. Cohen, "Fast image super-resolution based on in-place example regression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 1059–1066.

[37] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1646–1654.

[38] Y. Gao, H. Li, J. Dong, and G. Feng, "A deep convolutional network for medical image super-resolution," in *Proc. Chin. Autom. Congr. (CAC)*. Cham, Switzerland: Springer, Oct. 2017, pp. 184–199.

[39] C. Dong, C. C. Loy, and X. Tang, "Accelerating the super-resolution convolutional neural network," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 391–407.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[41] J. Kim, J. K. Lee, and K. M. Lee, "Deeply-recursive convolutional network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1637–1645.

[42] Y. Tai, J. Yang, and X. Liu, "Image super-resolution via deep recursive residual network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3147–3155.

[43] C. Ledig *et al.*, "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4681–4690.

[44] Y. Qi and H. Kobayashi, "On relation among time delay and signal strength based geolocation methods," in *Proc. IEEE Global Telecommun. Conf.*, vol. 7, Dec. 2003, pp. 4079–4083.

[45] Z. Yun and M. F. Iskander, "Ray tracing for radio propagation modeling: Principles and applications," *IEEE Access*, vol. 3, pp. 1089–1100, 2015.

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